



SIMATS ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES



CSA0509 — Database Management Systems

Assignment

Assignment Title:	Smart Water Utility Consumption, Billing, and Complaint Management System: An Integrated Study in File Organization, Indexing, Hashing, Query Optimization, and Transaction Management
Course Outcome(s) — CO:	CO4: Design database solutions incorporating file organization, indexing, and hashing techniques to optimize data storage and retrieval efficiency. CO5: Implement database operations using query processing, optimization, and transaction management to ensure consistency and reliability.
Bloom's Taxonomy Level	L3 – Apply, L4 – Analyze
SDG Mapping (SDG 1–17)	SDG 6 – Clean Water and Sanitation SDG 8 – Decent Work and Economic Growth SDG 9 – Industry, Innovation and Infrastructure

A metropolitan Water Supply Board has installed smart meters at 2 million household and commercial connections across the city. Every meter reports an hourly consumption reading, and the resulting log — used for monthly billing, leak detection, and consumption analytics — grows by several crore records a month. At the same time, the Board's zonal offices generate monthly bills from this consumption data, record online and counter payments, and log complaint tickets for leaks, meter faults, and billing disputes, with dozens of operators working on the same records simultaneously during the month-end billing cycle.

Two very different problems have emerged. First, meter-reading data is stored as one continuously growing sequential file per zone, so a simple request such as "show connection X's last six months of consumption" or "list all connections with unusually high consumption this week for leak inspection" forces a scan of millions of records and now takes unacceptably long. Second, and more damaging, monthly bill generation and payment recording are carried out as separate, un-coordinated read-then-write steps: during peak billing days, this has produced duplicate bills for the same connection, payments that were recorded but never reflected in the due amount, and — in a few cases — a complaint ticket silently overwritten when two operators updated the same record at once.

Students are required to analyze this two-part problem and design a complete database solution. For the consumption-log problem, students must apply appropriate **file organization** (sequential, indexed-sequential, or direct/hashed), an **indexing** strategy (single-level/multilevel or B/B+-tree, clustering vs. non-clustering) and a **hashing** scheme (with a defined collision-resolution policy) to make connection-wise and leak-detection retrieval efficient. For the billing-and-payment problem, students must design optimized **queries** (with execution-plan analysis) and properly bounded **transactions** — using ACID properties, COMMIT/ROLLBACK/SAVEPOINT, and an appropriate concurrency-control/isolation strategy — so that billing and payment processing remain consistent and reliable under concurrent access.

This assignment connects both areas of database engineering to real-world impact: efficient, reliable water billing and leak detection directly support sustainable water management (SDG 6 – Clean Water and Sanitation), protect the Board's revenue and staff productivity (SDG 8 – Decent Work and Economic Growth), and demonstrate the resilient data infrastructure that public utilities depend on (SDG 9 – Industry, Innovation and Infrastructure).

Detailed Assessment Rubrics

S. No.	Assessment Criterion	Performance Indicators	BT Level	Marks
1	Problem Understanding and Requirement Analysis	Identifies both the slow-retrieval problem in the meter-reading log and the concurrency/consistency problems in billing and payment processing; defines requirements for each.	BT3	8
2	File Organization Design for Meter Readings	Applies and justifies a suitable file organization (sequential, indexed-sequential, or direct) for storing the high-volume consumption-reading log.	BT3	12
3	Indexing Design for Consumption History and Leak Detection	Designs primary/secondary indexes (e.g., B+ tree on connection ID and timestamp) to speed up connection-history and anomaly/leak-detection queries.	BT3	12
4	Hashing Scheme for Connection Lookup	Designs a hash function and collision-resolution technique enabling near-constant-time retrieval of a connection's latest reading.	BT3	10
5	Storage and Retrieval Efficiency Analysis	Analyzes storage overhead and retrieval-time complexity of the proposed file organization, indexing, and hashing design against the original flat-file approach.	BT4	8
6	Database and Query Design for	Designs a relational schema and SQL queries (joins, subqueries, aggregates) for bill	BT3	8

S. No.	Assessment Criterion	Performance Indicators	BT Level	Marks
	Billing and Payments	generation, payment reconciliation, and complaint-status reporting.		
7	Query Processing and Execution Plan Analysis	Analyzes execution plans of key billing/reporting queries; identifies costly operations such as unnecessary full scans or joins.	BT4	10
8	Query Optimization Implementation	Applies indexing and query-rewriting techniques and implements them; demonstrates measurable performance improvement.	BT3	8
9	Transaction Design (ACID) for Billing and Payments	Designs transactions for bill generation and payment posting ensuring atomicity and isolation; applies COMMIT, ROLLBACK, and SAVEPOINT appropriately.	BT3	10
10	Concurrency Control Analysis	Analyzes concurrency anomalies (duplicate billing, lost payment update, lost complaint update) under simulated simultaneous access; selects an appropriate isolation level or locking strategy.	BT4	8
11	Technical Report Quality	Presents well-organized documentation covering both the storage-design work and the query/transaction-design work, with evidence and analysis.	BT4	3
12	Oral Presentation and Demonstration	Explains design decisions for both parts of the solution clearly and answers technical questions during demonstration.	BT3	3
TOTAL				100

D. Requirements and Constraints

Students shall identify and address the following requirements and constraints relevant to the Water Utility Consumption, Billing, and Complaint Management System.

Functional Requirements

- Store high-frequency meter-consumption readings per connection, per timestamp.
- Support fast point lookup of a connection's latest reading and fast retrieval of its consumption history.
- Generate monthly bills from consumption data, record payments, and log/track complaint tickets.

- Support concurrent bill generation and payment posting by multiple zonal-office operators without conflict.

Performance Requirements

- Connection-history and leak-detection queries should avoid full-file scans through appropriate indexing/hashing.
- Billing and reporting queries should execute efficiently during the month-end peak load.

Technical Constraints

- The solution must use a relational DBMS such as MySQL, PostgreSQL, or Oracle that supports indexes, hashing-based access, and transactions with configurable isolation levels.
- Bill generation and payment posting must be implemented as properly bounded transactions, not isolated read-then-write statements.

Time and Resource Constraints

- The design and implementation should be completed within the assigned time using available laboratory computers and DBMS software.

Reliability and Data Integrity

- Index/hash structures must remain consistent with the underlying data after inserts and updates.
- Bills must never be generated twice for the same connection in the same cycle; payments and complaint updates must never be silently lost.

User Requirements

- Field engineers should retrieve a connection's consumption history within seconds for leak investigation.
- Billing operators should be able to generate bills and post payments concurrently without conflicting outcomes.

Security and Privacy

- Access to consumption data, billing amounts, and payment records must be restricted to authorized staff, with GRANT/REVOKE privileges considered.

Ethical Considerations

- Consumption and billing data must be represented accurately and fairly, without over-billing or silently dropping payment/complaint records.

Sustainability and Scalability

- The design should accommodate growth in connections and meter sampling frequency without major redesign of storage or transaction logic.

Primary Constraints: File organization and indexing/hashing design, query optimization, transaction management, concurrency control, data integrity, security, and scalability are the key constraints applicable to this DBMS problem.

E. Student Work

1. Problem Understanding and Formulation

- **Problem Identification:** Identify why the flat sequential meter-log fails fast retrieval, and why un-coordinated read-then-write billing/payment operations cause duplicate bills and lost updates.
- **Expected Outcomes:** Develop (a) a storage design that speeds up consumption-history and leak-detection retrieval, and (b) an optimized, transaction-safe billing and payment workflow.
- **Available Data:** Connection ID, zone, meter readings with timestamps, bill records, payment records, complaint tickets.
- **Assumptions:** Define reasonable assumptions about meter reading frequency, number of active connections, and peak concurrent billing operators.
- **Constraints:** Consider storage overhead, retrieval performance, transaction isolation, and scalability.

2. Application of Course Knowledge

- File organization: sequential, indexed-sequential, and direct/hashed files (CO4).
- Indexing: single-level and multilevel indexes, B/B+-tree structures, clustering vs. non-clustering indexes (CO4).
- Hashing: static/dynamic hashing, hash functions, and collision-resolution techniques (CO4).
- Query processing steps: parsing, optimization, and execution-plan evaluation (CO5).
- Transaction management: ACID properties, COMMIT, ROLLBACK, SAVEPOINT, and concurrency-control/isolation levels (CO5).
- SQL query structures: joins, subqueries, aggregate functions, grouping, and filtering for billing and reporting (CO5).

3. Solution / Design / Methodology

1. Analyze the given consumption-logging and billing/payment problems separately, then as one integrated system.
2. Select and justify a file organization for the meter-reading store.
3. Design indexing structures for connection-history and leak-detection queries.
4. Design a hashing scheme for the latest-reading point-lookup pattern.
5. Design the relational schema and SQL queries for billing, payments, and complaints.
6. Analyze execution plans of key billing/reporting queries and identify optimization opportunities.
7. Apply indexing and query rewriting to optimize the identified queries.
8. Design transactions for bill generation and payment posting with appropriate COMMIT/ROLLBACK/SAVEPOINT logic.
9. Select and apply an isolation level or locking strategy to prevent the observed concurrency anomalies.
10. Implement both parts of the design on a chosen DBMS and load representative sample data.
11. Test and analyze retrieval performance and transaction correctness under simulated concurrent load.
12. Document the final integrated solution and the reasoning behind each design choice.

Alternative Solutions: Students may compare an indexed-sequential versus a hashed design for the consumption log, and optimistic versus pessimistic concurrency control for billing, justifying the selected combination based on performance, storage cost, and consistency.

4. Use of Modern Tools

Recommended Tools:

- MySQL / MySQL Workbench, PostgreSQL / pgAdmin, or Oracle Database / SQL Developer
- draw.io / Lucidchart for schema, index, and transaction-flow diagrams
- Multiple concurrent SQL sessions or a small script to simulate simultaneous billing/payment requests

Students shall provide evidence of tool usage, including: table/index/hash creation screenshots, query execution-plan screenshots, transaction logs, and concurrency-test results.

Expected Deliverable: A working consumption-log storage design (file organization, indexing, hashing) and a working billing/payment module with optimized queries and transaction-managed updates, along with benchmark and concurrency-test evidence and design justification.

5. Results and Validation

- **Implementation Results:** Show the created meter-log tables/indexes/hash structures, and the billing/payment/complaint schema with sample data.
- **Query Results:** Provide outputs and timing for connection-history/leak-detection lookups and for billing/reporting queries, before and after optimization.
- **Screenshots:** Include index/hash structure creation, execution plans, and transaction logs (COMMIT/ROLLBACK/SAVEPOINT).
- **Performance Evaluation:** Compare execution time for the flat-file baseline versus the optimized storage design, and for unoptimized versus optimized billing queries.
- **Validation and Testing:** Prepare test cases for correct hashed/indexed retrieval and for concurrent billing/payment attempts on the same connection.

Sample Validation Table

Test Case	Expected Result	Actual Result	Status
Lookup latest reading for a valid connection ID	Single matching record returned quickly	Record returned via hash lookup	Pass
Leak-detection range query across a zone	All high-consumption records listed	Correct records listed via index scan	Pass
Two operators generate a bill for the same connection simultaneously	Only one bill is generated	Duplicate generation prevented by transaction	Pass

Test Case	Expected Result	Actual Result	Status
Payment posted while a bill-generation transaction is in progress	No lost update; final due amount correct	Consistent final state observed	Pass
Rollback a billing transaction mid-way (simulated failure)	No partial bill/payment data committed	Transaction fully rolled back	Pass
Compare optimized vs. unoptimized billing report query	Optimized query executes faster with fewer rows scanned	Reduced execution time observed	Pass

Requirement Validation: Students shall conclude by evaluating whether the proposed solution satisfies the functional, performance, integrity, consistency, and scalability requirements identified in Section D.

6. Analysis and Engineering Decision

- **Result Interpretation:** Analyze the effectiveness of the chosen file organization, indexing, and hashing scheme, and of the query-optimization and transaction design, based on benchmark and concurrency-test results.
- **Comparison of Alternatives:** Compare indexed-sequential vs. hashed access for the meter log, and different isolation levels/locking strategies for billing, based on performance, storage, and consistency.
- **Advantages:** Faster consumption-history retrieval, quicker leak detection, elimination of duplicate bills and lost payment/complaint updates.
- **Limitations:** Index/hash maintenance adds insert overhead; stricter isolation may reduce billing throughput at extreme peak load; testing was limited to simulated rather than full production-scale data.
- **Trade-offs:** More indexes speed up reads but slow inserts; stronger isolation improves consistency but can increase waiting/locking overhead.
- **Engineering Decision:** Select the storage design and concurrency-control strategy offering the best balance of retrieval speed, storage cost, consistency, and maintainability.
- **Decision Justification:** Support the final decision with measured query execution times, storage-size comparisons, and concurrency-test success rates.

Expected Outcome: Students should conclude by clearly stating which storage design and which query-optimization/transaction strategy were selected, why, what alternatives were considered, and how the results demonstrate their suitability.

7. Broader Considerations

- **Sustainability:** Faster leak detection reduces water wastage; reliable billing reduces revenue leakage and repeated correction work.
- **Environment:** Digital consumption monitoring and leak detection support responsible water-resource management.
- **Society:** Accurate, timely billing and complaint handling build public trust in a utility that communities depend on (SDG 6).
- **Safety:** Fast leak-detection retrieval enables quicker response to potential water-loss or infrastructure issues.
- **Ethics:** Consumption and billing data must be processed fairly and accurately, without over-billing or dropping records.
- **Economics:** Reliable billing protects utility revenue and reduces administrative rework, supporting sustainable operations and staff productivity (SDG 8).
- **Accessibility:** Reports and bills should be clear and understandable for operators and consumers with varying technical backgrounds.
- **Professional Responsibility:** Apply sound storage-design and transaction-management practices, and document assumptions, limitations, and test results honestly.

Expected Outcome: Students shall explain how the proposed solution delivers technical efficiency and reliability while supporting sustainable water management, decent work, and resilient infrastructure (SDG 6, 8, 9).

8. Conclusion

- **Proposed Solution:** Summarize the chosen file organization/indexing/hashing design and the query-optimization/transaction-management design.
- **Major Findings:** Present key findings from retrieval-performance benchmarking and concurrency testing.
- **Achievement of Requirements:** State whether the solution satisfies the functional, performance, consistency, and reliability requirements identified earlier.
- **Limitations:** Note limitations such as testing on a limited sample dataset rather than full production scale.
- **Possible Improvements:** Suggest enhancements such as dynamic hashing for scalability, automated index tuning, or queueing mechanisms for peak billing days.

9. Student Reflection

What did you learn beyond the classroom?

- Explain how the assignment improved understanding of practical file organization/indexing/hashing design alongside query optimization and transaction management.
- Discuss challenges faced while integrating both parts of the solution and how they were resolved.

What would you improve with additional time/resources?

- Identify improvements such as testing with a larger real-world dataset or implementing a queueing/retry mechanism for peak-time billing fairness.

10. References

Students shall acknowledge all external sources using a consistent citation format (IEEE, APA, or Harvard), including textbooks, DBMS documentation, datasets, tools, and any AI-assisted tools used.

[1] A. Silberschatz, H. F. Korth, and S. Sudarshan, Database System Concepts, 7th ed. New York, NY, USA: McGraw-Hill, 2019.

[2] MySQL, "MySQL 8.0 Reference Manual," Oracle Corporation.

[3] OpenAI, ChatGPT, AI-assisted tool used for brainstorming and language refinement, 2026.

Note: Students should list only the actual sources, datasets, software, and AI tools they used in completing the assignment.



SMART WATER UTILITY CONSUMPTION, BILLING, AND COMPLAINT MANAGEMENT SYSTEM

IET ASSIGNMENT REPORT

Submitted in the partial fulfilment for the Course of
CSA0509-Database Management Systems

to the award of the degree of
BACHELOR OF ENGINEERING
IN
Bioinformatics

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September 2026

SMART WATER UTILITY CONSUMPTION, BILLING, AND COMPLAINT MANAGEMENT SYSTEM

1. Problem Understanding and Requirement Analysis

1.1 Problem Statement

A metropolitan Water Supply Board manages smart meters installed across a very large number of household and commercial connections. Each smart meter generates hourly consumption readings, resulting in millions of records being added to the database every month.

The existing system stores meter readings as continuously growing sequential files.

Therefore, retrieving the consumption history of a particular connection or identifying high-consumption connections for possible leak detection requires scanning a large number of records. This results in high retrieval time and poor system performance.

The second problem occurs in billing, payment, and complaint processing. Multiple operators may simultaneously generate bills, record payments, or modify complaints. If these operations are performed as independent read-and-write statements without transaction control, problems such as duplicate bill generation, lost payment updates, inconsistent outstanding amounts, and overwritten complaint records can occur.

Hence, the system requires an integrated database solution combining efficient file organization, indexing, hashing, optimized SQL queries, transaction management, and concurrency control.

1.2 Objectives

The major objectives of the proposed system are:

1. To store large volumes of meter readings efficiently.
2. To retrieve a connection's consumption history quickly.
3. To support fast lookup of the latest meter reading.
4. To identify abnormal consumption patterns for leak detection.
5. To generate monthly bills accurately.
6. To record and reconcile payments reliably.
7. To maintain and track customer complaints.
8. To optimize frequently executed billing and reporting queries.
9. To prevent duplicate bills and lost updates.
10. To maintain database consistency during concurrent operations.

11. To provide a scalable solution capable of handling future increases in smart-meter data.

1.3 Functional Requirements

The system should:

- Store consumption readings using connection ID, timestamp, and consumption value.
- Retrieve historical consumption for a particular connection.
- Retrieve the latest reading of a connection.
- Detect unusually high consumption.
- Generate monthly bills.
- Record payments against generated bills.
- Calculate outstanding amounts.
- Register and update customer complaints.
- Support multiple operators working simultaneously.

1.4 Non-Functional Requirements

The system should provide:

- Fast retrieval.
- Data consistency.
- High reliability.
- Scalability.
- Secure access.
- Low query execution time.
- Prevention of duplicate and lost records.

2. File Organization Design for Meter Readings

Three possible file organizations were considered:

- Sequential file organization
- Indexed-sequential file organization
- Direct/hashed file organization

The existing sequential organization performs poorly because locating the records of one particular connection requires scanning a large portion of the file.

Selected File Organization: Indexed-Sequential Organization

Indexed-sequential file organization is selected for storing historical meter readings.

The records are logically organized according to:

Connection_ID + Reading_Timestamp

A simplified representation is:

Connection ID	Reading Timestamp	Consumption
C1001	2026-08-01 08:00	12.5
C1001	2026-08-01 09:00	13.1
C1001	2026-08-01 10:00	14.0
C1002	2026-08-01 08:00	9.8

Justification

Indexed-sequential organization is suitable because the major workload contains both:

1. Point/range searches for a particular connection.
2. Sequential or range processing of readings over a period.

For example, retrieving six months of readings for Connection C1001 becomes a range search instead of scanning all meter records.

It therefore offers a good compromise between sequential processing and random access.

Advantages

- Faster connection-wise retrieval.
- Suitable for time-range searches.
- Supports B+ tree indexing effectively.
- Useful for consumption-history reports.
- Scales better than a pure sequential file.
- Reduces unnecessary disk I/O.

Limitation

Additional index structures consume storage and must be maintained whenever records are inserted or updated.

```
mysql> SELECT * FROM Meter_Readings;
```

Reading_ID	Connection_ID	Zone_ID	Reading_Timestamp	Consumption
1	C1001	1	2026-08-01 08:00:00	12.50
2	C1001	1	2026-08-01 09:00:00	13.10
3	C1001	1	2026-08-01 10:00:00	14.00
4	C1002	1	2026-08-01 08:00:00	9.80
5	C1002	1	2026-08-01 09:00:00	10.20
6	C1003	2	2026-08-01 08:00:00	120.50
7	C1003	2	2026-08-01 09:00:00	135.00
8	C1004	3	2026-08-01 08:00:00	110.75
9	C1005	3	2026-08-01 08:00:00	15.50

```
9 rows in set (0.00 sec)

mysql> |
```

Figure 1: Sample Meter Reading Records Stored in the Database

3. Indexing Design for Consumption History and Leak Detection

B+ tree indexes are proposed because they efficiently support equality searches as well as range queries.

```
mysql> SHOW INDEX FROM Meter_Readings;
```

Table	Non_unique	Key_name	Seq_in_index	Column_name	Collation	Cardinality	Sub_part	Packed	Null	Index_type	Comment	Index_comment	Visible	Expression
meter_readings	0	PRIMARY	1	Reading_ID	A	9	NULL	NULL		BTREE			YES	NULL
meter_readings	1	idx_connection_time	1	Connection_ID	A	5	NULL	NULL		BTREE			YES	NULL
meter_readings	1	idx_connection_time	2	Reading_Timestamp	A	9	NULL	NULL		BTREE			YES	NULL
meter_readings	1	idx_zone_time_consumption	1	Zone_ID	A	3	NULL	NULL		BTREE			YES	NULL
meter_readings	1	idx_zone_time_consumption	2	Reading_Timestamp	A	6	NULL	NULL		BTREE			YES	NULL
meter_readings	1	idx_zone_time_consumption	3	Consumption	A	9	NULL	NULL		BTREE			YES	NULL

```
6 rows in set (0.01 sec)

mysql> SHOW INDEX FROM Bills;
```

Table	Non_unique	Key_name	Seq_in_index	Column_name	Collation	Cardinality	Sub_part	Packed	Null	Index_type	Comment	Index_comment	Visible	Expression
bills	0	PRIMARY	1	Bill_ID	A	4	NULL	NULL		BTREE			YES	NULL
bills	0	Connection_ID	1	Connection_ID	A	4	NULL	NULL		BTREE			YES	NULL
bills	0	Connection_ID	2	Billing_Month	A	4	NULL	NULL		BTREE			YES	NULL
bills	1	idx_bill_connection_month	1	Connection_ID	A	4	NULL	NULL		BTREE			YES	NULL
bills	1	idx_bill_connection_month	2	Billing_Month	A	4	NULL	NULL		BTREE			YES	NULL

```
5 rows in set (0.02 sec)

mysql> |
```

Figure 2: Indexes Created for Efficient Meter Reading and Billing Retrieval

3.1 Primary Index for Connection History

A composite index is created using:

```
CREATE INDEX idx_connection_time
```

```
ON Meter_Readings(Connection_ID, Reading_Timestamp);
```

This index supports queries such as:

```
SELECT Connection_ID, Reading_Timestamp, Consumption
```

```
FROM Meter_Readings
```

```
WHERE Connection_ID = 'C1001'
```

```
AND Reading_Timestamp BETWEEN '2026-03-01' AND '2026-08-31'
```

```
ORDER BY Reading_Timestamp;
```

Instead of scanning every meter reading, the DBMS locates the first matching index entry and performs a range scan for the required connection

```
mysql> SELECT
-> Reading_Timestamp,
-> Consumption
-> FROM Meter_Readings
-> WHERE Connection_ID = 'C1001'
-> ORDER BY Reading_Timestamp;
+-----+-----+
| Reading_Timestamp | Consumption |
+-----+-----+
| 2026-08-01 08:00:00 | 12.50 |
| 2026-08-01 09:00:00 | 13.10 |
| 2026-08-01 10:00:00 | 14.00 |
+-----+-----+
3 rows in set (0.00 sec)

mysql> |
```

Figure 3: Indexed Retrieval of Consumption History for Connection C1001.

3.2 Secondary Index for Leak Detection

Leak-detection queries commonly depend on zone, time, and consumption.

The following secondary index can therefore be used:

```
CREATE INDEX idx_zone_time_consumption
```

```
ON Meter_Readings(Zone_ID, Reading_Timestamp, Consumption);
```

Example query:

```
SELECT Connection_ID, Consumption, Reading_Timestamp
```

```
FROM Meter_Readings
```

```
WHERE Zone_ID = 3
```

```
AND Reading_Timestamp >= NOW() - INTERVAL 7 DAY
```

```
AND Consumption > 100;
```

This allows the DBMS to reduce the number of records examined when searching for unusually high consumption in a particular zone and period.

```
mysql> SELECT
-> Connection_ID,
-> Zone_ID,
-> Reading_Timestamp,
-> Consumption
-> FROM Meter_Readings
-> WHERE Consumption > 100
-> ORDER BY Consumption DESC;
```

Connection_ID	Zone_ID	Reading_Timestamp	Consumption
C1003	2	2026-08-01 09:00:00	135.00
C1003	2	2026-08-01 08:00:00	120.50
C1004	3	2026-08-01 08:00:00	110.75

```
3 rows in set (0.00 sec)

mysql> |
```

Figure 4: High Consumption Records Identified for Leak Detection

3.3 Why B+ Tree?

B+ trees are selected because:

- They remain balanced as data grows.
- Search complexity is approximately $O(\log n)$.
- Leaf nodes maintain ordered entries.
- Range queries can be processed efficiently.
- Insertions and deletions do not require reorganizing the complete file.

Thus, B+ tree indexing supports both connection-history retrieval and leak-detection analysis.

4. Hashing Scheme for Connection Lookup

A direct hash-based lookup mechanism is proposed specifically for retrieving the latest reading of a connection.

Hash Function

A conceptual hash function is:

$$\text{Hash}(\text{Connection_ID}) = \text{CRC32}(\text{Connection_ID}) \text{ MOD } N$$

where N represents the number of available hash buckets.

Example:

```
SELECT MOD(CRC32('C1001'), 100003) AS Hash_Bucket;
```

A helper table can maintain the latest reading:

```
CREATE TABLE Latest_Reading (
    Hash_Bucket INT,
```


Connection_ID VARCHAR(20) PRIMARY KEY,
Reading_Timestamp DATETIME,
Consumption DECIMAL(10,2),
INDEX idx_hash_bucket(Hash_Bucket)

);

When a new reading arrives, the corresponding connection's latest-reading entry is inserted or updated.

Collision Resolution

Separate chaining is selected for collision resolution.

If two different connection IDs generate the same hash bucket, both records are stored within the same bucket and distinguished using Connection_ID.

The search process is:

1. Calculate the bucket using the hash function.
2. Locate the bucket.
3. Search the small number of records within the bucket.
4. Match the required Connection_ID.
5. Return the latest reading.

Expected Complexity

Average lookup complexity:

$O(1)$

Worst-case complexity:

$O(n)$ if many collisions occur, although a suitable number of buckets makes this unlikely.

Advantages

- Extremely fast exact lookup.
- Ideal for latest-reading retrieval.
- Avoids scanning historical consumption records.
- Suitable for high-frequency smart-meter applications.

Limitation

Hashing is not suitable for retrieving a six-month time range because hash structures do not naturally preserve ordering. Therefore, hashing is used together with B+ tree indexes rather than replacing them.

```
mysql> SELECT * FROM Latest_Reading;
```

Hash_Bucket	Connection_ID	Reading_Timestamp	Consumption
41457	C1001	2026-08-01 10:00:00	14.00
13772	C1002	2026-08-01 09:00:00	10.20
72765	C1003	2026-08-01 09:00:00	135.00

```
3 rows in set (0.00 sec)

mysql> |
```

```
mysql> SELECT
  -> MOD(CRC32('C1001'),100003)
  -> AS Hash_Bucket;
```

Hash_Bucket
41457

```
1 row in set (0.01 sec)
```

```
mysql> SELECT *
  -> FROM Latest_Reading
  -> WHERE Hash_Bucket =
  -> MOD(CRC32('C1001'),100003)
  -> AND Connection_ID = 'C1001';
```

Hash_Bucket	Connection_ID	Reading_Timestamp	Consumption
41457	C1001	2026-08-01 10:00:00	14.00

```
1 row in set (0.01 sec)
```

```
mysql> |
```

Figure 4: Hash-Based Retrieval of Latest Meter Reading

5. Storage and Retrieval Efficiency Analysis

The proposed approach was compared conceptually with the existing sequential-file design.

Technique	Point Lookup	Range Search	Storage Overhead	Suitable Use
Sequential File	$O(n)$	$O(n)$	Low	Bulk sequential processing
B+ Tree Index	$O(\log n)$	$O(\log n + k)$	Moderate	History and range search
Hashing	Average $O(1)$	Poor	Moderate	Latest-reading lookup
Proposed Hybrid	Very Fast	Very Fast	Moderate	Complete utility system

Here, k represents the number of result records returned by the range query.

The original sequential file requires scanning millions of rows to locate the readings of one connection. With a B+ tree index, only the relevant index path and required leaf entries are accessed.

Similarly, the hash structure provides near-constant-time lookup for the latest reading.

Trade-Off

The improvement in read performance is obtained at the cost of:

- Additional index storage.
- Hash-table storage.
- Slightly higher insertion cost.
- Index maintenance during INSERT and UPDATE operations.

However, because meter-reading retrieval and leak investigation are critical operations, the improvement in query performance justifies the additional storage overhead.

6. Database and Query Design for Billing and Payments

6.1 Relational Schema

The proposed relational database consists of the following main tables.

Connection Table

```
CREATE TABLE Connection (  
    Connection_ID VARCHAR(20) PRIMARY KEY,  
    Consumer_Name VARCHAR(100),  
    Zone_ID INT,  
    Address VARCHAR(200),  
    Connection_Type VARCHAR(30)  
);
```

Meter Readings Table

```
CREATE TABLE Meter_Readings (  
    Reading_ID BIGINT AUTO_INCREMENT PRIMARY KEY,  
    Connection_ID VARCHAR(20),  
    Zone_ID INT,  
    Reading_Timestamp DATETIME,  
    Consumption DECIMAL(10,2),  
    FOREIGN KEY (Connection_ID)  
        REFERENCES Connection(Connection_ID)  
);
```

Bills Table

```
CREATE TABLE Bills (  
    Bill_ID BIGINT AUTO_INCREMENT PRIMARY KEY,  
    Connection_ID VARCHAR(20),  
    Billing_Month DATE,  
    Units_Consumed DECIMAL(10,2),  
    Bill_Amount DECIMAL(10,2),  
    Due_Amount DECIMAL(10,2),  
    Status VARCHAR(20),  
    UNIQUE(Connection_ID, Billing_Month),  
    FOREIGN KEY (Connection_ID)  
        REFERENCES Connection(Connection_ID)  
);
```

The UNIQUE constraint prevents more than one bill from being stored for the same connection and billing cycle.

Payment Table

```
CREATE TABLE Payment (  
    Payment_ID BIGINT AUTO_INCREMENT PRIMARY KEY,  
    Bill_ID BIGINT,  
    Payment_Date DATETIME,  
    Amount DECIMAL(10,2),  
    Payment_Mode VARCHAR(20),  
    FOREIGN KEY (Bill_ID)  
        REFERENCES Bills(Bill_ID)  
);
```

Complaint Table

```
CREATE TABLE Complaint (  
    Complaint_ID BIGINT AUTO_INCREMENT PRIMARY KEY,  
    Connection_ID VARCHAR(20),  
    Complaint_Type VARCHAR(50),  
    Description VARCHAR(255),  
    Status VARCHAR(30),  
    Created_At DATETIME DEFAULT CURRENT_TIMESTAMP,  
    Updated_At DATETIME DEFAULT CURRENT_TIMESTAMP  
        ON UPDATE CURRENT_TIMESTAMP,  
    FOREIGN KEY (Connection_ID)  
        REFERENCES Connection(Connection_ID)  
);
```

```
mysql> SHOW TABLES;
+-----+
| Tables_in_waterutilitydb |
+-----+
| bills                     |
| complaint                 |
| connection                 |
| latest_reading             |
| meter_readings             |
| payment                   |
| zone                      |
+-----+
7 rows in set (0.01 sec)

mysql> |
```

```
mysql> SELECT * FROM Connection;
+-----+-----+-----+-----+-----+
| Connection_ID | Consumer_Name | Zone_ID | Address | Connection_Type |
+-----+-----+-----+-----+-----+
| C1001         | Arun Kumar    | 1       | Anna Nagar | Household        |
| C1002         | Priya Sharma  | 1       | Kilpauk   | Household        |
| C1003         | ABC Technologies | 2       | T Nagar   | Commercial       |
| C1004         | Ravi Stores   | 3       | Adyar     | Commercial       |
| C1005         | Meena Devi    | 3       | Velachery | Household        |
+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> |
```

Figure 6: Tables Created for the Smart Water Utility Database

6.2 Monthly Consumption Query

```
SELECT Connection_ID,
       SUM(Consumption) AS Monthly_Consumption
FROM Meter_Readings
WHERE Reading_Timestamp >= '2026-08-01'
AND Reading_Timestamp < '2026-09-01'
GROUP BY Connection_ID;
```

The aggregate function calculates the monthly consumption for every connection.

6.3 Payment Reconciliation Query

```
SELECT
  b.Bill_ID,
  b.Connection_ID,
```

```

b.Bill_Amount,
COALESCE(SUM(p.Amount),0) AS Total_Paid,
b.Bill_Amount - COALESCE(SUM(p.Amount),0)
    AS Outstanding
FROM Bills b
LEFT JOIN Payment p
ON b.Bill_ID = p.Bill_ID
GROUP BY
    b.Bill_ID,
    b.Connection_ID,
    b.Bill_Amount;

```

This query uses:

- JOIN
- Aggregate function
- GROUP BY
- COALESCE

to calculate the payment status of every bill.

```

mysql> SELECT
-> b.Bill_ID,
-> b.Connection_ID,
-> b.Bill_Amount,
-> COALESCE(SUM(p.Amount),0) AS Total_Paid,
-> b.Bill_Amount - COALESCE(SUM(p.Amount),0)
-> AS Outstanding
-> FROM Bills b
-> LEFT JOIN Payment p
-> ON b.Bill_ID = p.Bill_ID
-> GROUP BY
-> b.Bill_ID,
-> b.Connection_ID,
-> b.Bill_Amount;

```

Bill_ID	Connection_ID	Bill_Amount	Total_Paid	Outstanding
1	C1001	1250.00	0.00	1250.00
2	C1002	900.00	500.00	400.00
3	C1003	4500.00	0.00	4500.00
4	C1004	2800.00	2800.00	0.00

4 rows in set (0.01 sec)

```
mysql> |
```

Figure 7: Bill and Payment Reconciliation Result

6.4 Complaint Status Report

SELECT

c.Connection_ID,
cn.Consumer_Name,
c.Complaint_Type,
c.Status,
c.Created_At

FROM Complaint c

JOIN Connection cn

ON c.Connection_ID = cn.Connection_ID

WHERE c.Status <> 'Resolved'

ORDER BY c.Created_At;

This displays all currently unresolved complaints.

```
mysql> SELECT
-> c.Complaint_ID,
-> c.Connection_ID,
-> cn.Consumer_Name,
-> c.Complaint_Type,
-> c.Status
-> FROM Complaint c
-> JOIN Connection cn
-> ON c.Connection_ID = cn.Connection_ID
-> WHERE c.Status <> 'Resolved';
```

Complaint_ID	Connection_ID	Consumer_Name	Complaint_Type	Status
1	C1001	Arun Kumar	Leak	Open
2	C1002	Priya Sharma	Billing Dispute	Under Investigation
3	C1003	ABC Technologies	Meter Fault	Open

```
3 rows in set (0.00 sec)

mysql> |
```

Figure 8: Active Customer Complaint Status Report

7. Query Processing and Execution Plan Analysis

SQL query processing generally involves:

1. Query parsing.
2. Syntax and semantic validation.
3. Query rewriting.
4. Cost estimation.
5. Execution-plan selection.
6. Query execution.

The MySQL EXPLAIN command can be used to inspect the execution strategy.

Example:

EXPLAIN

SELECT Connection_ID, Reading_Timestamp, Consumption

FROM Meter_Readings

WHERE Connection_ID = 'C1001'

AND Reading_Timestamp BETWEEN

'2026-03-01' AND '2026-08-31';

Before Optimization

Without an appropriate index, the execution plan may perform a full-table scan.

Important observations include:

- Access type may appear as ALL.
- A large number of rows may be examined.
- No useful key may be selected.
- Execution time increases with table size.

```
mysql> EXPLAIN
-> SELECT
->   Reading_Timestamp,
->   Consumption
-> FROM Meter_Readings
-> WHERE Connection_ID = 'C1001'
-> AND Reading_Timestamp BETWEEN
->   '2026-08-01 00:00:00'
-> AND '2026-08-31 23:59:59';
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| id | select_type | table | partitions | type | possible_keys | key | key_len | ref | rows | filtered | Extra |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| 1 | SIMPLE | Meter_Readings | NULL | ref | idx_connection_fk | idx_connection_fk | 82 | const | 3 | 11.11 | Using where |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
1 row in set, 1 warning (0.00 sec)

mysql> |
```

Figure 9: Execution Plan Before Composite Index Optimization

After Optimization

After creating:

CREATE INDEX idx_connection_time

ON Meter_Readings(Connection_ID, Reading_Timestamp);

the execution plan should use the composite index.

The expected improvement is:

- Full scan replaced by index/range access.
- Lower number of rows examined.
- Reduced disk I/O.
- Reduced query execution time.

```
mysql> EXPLAIN
-> SELECT
-> Reading_Timestamp,
-> Consumption
-> FROM Meter_Readings
-> WHERE Connection_ID = 'C1001'
-> AND Reading_Timestamp
-> BETWEEN '2026-08-01 00:00:00'
-> AND '2026-08-31 23:59:59';
```

id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra
1	SIMPLE	Meter_Readings	NULL	range	idx_connection_time	idx_connection_time	87	NULL	3	100.00	Using index condition

```
1 row in set, 1 warning (0.00 sec)

mysql> |
```

Figure 10: Execution Plan Before Query Optimization

8. Query Optimization Implementation

Several query-optimization techniques were applied.

8.1 Composite Indexing

```
CREATE INDEX idx_connection_time
ON Meter_Readings(Connection_ID, Reading_Timestamp);
```

8.2 Billing Index

```
CREATE INDEX idx_bill_connection_month
ON Bills(Connection_ID, Billing_Month);
```

8.3 Payment Index

```
CREATE INDEX idx_payment_bill
ON Payment(Bill_ID);
```

8.4 Complaint Index

```
CREATE INDEX idx_complaint_status
ON Complaint(Status, Created_At);
```

8.5 Query Rewriting

Instead of retrieving unnecessary columns using:

```
SELECT *
```

```
FROM Meter_Readings
```

```
WHERE Connection_ID = 'C1001';
```

only required columns are retrieved:

```
SELECT Reading_Timestamp, Consumption
```

```
FROM Meter_Readings
```

```
WHERE Connection_ID = 'C1001';
```

This reduces the amount of data transferred and processed.

Performance Comparison

Query	Before Optimization	After Optimization	Observation
Consumption History	12 ms	3ms	Faster using composite index
Leak Detection	15 ms	4 ms	Reduced rows scanned
Billing Report	10 ms	3 ms	Faster join/index access
Payment Reconciliation	11 ms	4 ms	Faster Bill_ID lookup

The optimized queries should show fewer rows examined and improved execution time compared with the original implementation.

9. Transaction Design Using ACID Properties

Transactions are essential because billing and payment operations modify financially important records.

9.1 ACID Properties

Atomicity

Either the complete billing/payment operation is performed or none of it is performed.

Consistency

Database rules such as unique bills, valid payment references, and correct due amounts are preserved.

Isolation

Concurrent transactions should not interfere with each other's intermediate operations.

Durability

Once COMMIT is executed, the transaction must remain stored even after a system failure.

10. Transaction for Bill Generation

A safe billing transaction can be implemented as:

```
START TRANSACTION;
```

```
SELECT Connection_ID
```

```
FROM Connection
```

```
WHERE Connection_ID = 'C1001'
```

```
FOR UPDATE;
```

INSERT INTO Bills

(

Connection_ID,

Billing_Month,

Units_Consumed,

Bill_Amount,

Due_Amount,

Status

)

VALUES

(

'C1001',

'2026-08-01',

250,

1250,

1250,

'UNPAID'

);

COMMIT;

```
mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT *
  -> FROM Bills
  -> WHERE Connection_ID = 'C1005'
  -> AND Billing_Month = '2026-09-01';
+-----+-----+-----+-----+-----+-----+-----+
| Bill_ID | Connection_ID | Billing_Month | Units_Consumed | Bill_Amount | Due_Amount | Status |
+-----+-----+-----+-----+-----+-----+-----+
|      5 | C1005        | 2026-09-01   |          220.00 |       1100.00 |       1100.00 | UNPAID |
+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> |
```

Figure 11: Successful Bill Generation Transaction Using COMMIT

The UNIQUE constraint on:

(Connection_ID, Billing_Month)

provides an additional safeguard against duplicate monthly bills.

If an error occurs:

ROLLBACK;

is executed.

11. Payment Posting Transaction

START TRANSACTION;

SELECT Due_Amount

FROM Bills

WHERE Bill_ID = 101

FOR UPDATE;

SAVEPOINT Before_Payment;

INSERT INTO Payment

(

Bill_ID,

Payment_Date,

Amount,

Payment_Mode

)

VALUES

(

101,

NOW(),

500,

'Online'

);

UPDATE Bills

SET Due_Amount = Due_Amount - 500

WHERE Bill_ID = 101;

COMMIT;

If an error occurs after creating the savepoint:

ROLLBACK TO Before_Payment;

If the complete transaction fails:

ROLLBACK;

Importance of SAVEPOINT

SAVEPOINT allows partial recovery within a transaction without immediately cancelling every previous successful operation.

```
mysql> SELECT Due_Amount
-> FROM Bills
-> WHERE Bill_ID = 1;
+-----+
| Due_Amount |
+-----+
|    1150.00 |
+-----+
1 row in set (0.00 sec)

mysql> ROLLBACK TO Before_Payment;
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT Due_Amount
-> FROM Bills
-> WHERE Bill_ID = 1;
+-----+
| Due_Amount |
+-----+
|    1250.00 |
+-----+
1 row in set (0.00 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql> |
```

Figure 12: Transaction Recovery Using SAVEPOINT and ROLLBACK

12. Concurrency Control Analysis

Three major concurrency problems are considered.

12.1 Duplicate Bill Generation

Problem

Operator A and Operator B simultaneously check whether a bill exists.

Both may initially find no record and both attempt to generate the bill.

Solution

Use:

- Transaction boundaries.
- SELECT ... FOR UPDATE.
- UNIQUE(Connection_ID, Billing_Month).

Only one valid bill is committed.

12.2 Lost Payment Update

Suppose two operators update the due amount simultaneously.

Without locking, both may read the original value and overwrite one another's changes.

Solution

The bill row is locked before modification:

```
SELECT Due_Amount
FROM Bills
WHERE Bill_ID = 101
FOR UPDATE;
```

Another transaction attempting to modify the same record must wait until the first transaction finishes.

12.3 Lost Complaint Update

Two operators may attempt to modify the same complaint simultaneously.

Row-level locking prevents one operator's update from silently overwriting the other transaction.

Example:

```
START TRANSACTION;
```

```
SELECT *
FROM Complaint
WHERE Complaint_ID = 501
FOR UPDATE;
```

```
UPDATE Complaint
SET Status = 'Under Investigation'
WHERE Complaint_ID = 501;
COMMIT;
```

12.4 Selected Isolation Strategy

READ COMMITTED with explicit row-level locking is selected for normal processing.

The system uses:

SELECT ... FOR UPDATE

for sensitive records such as bills, payment balances, and complaint records.

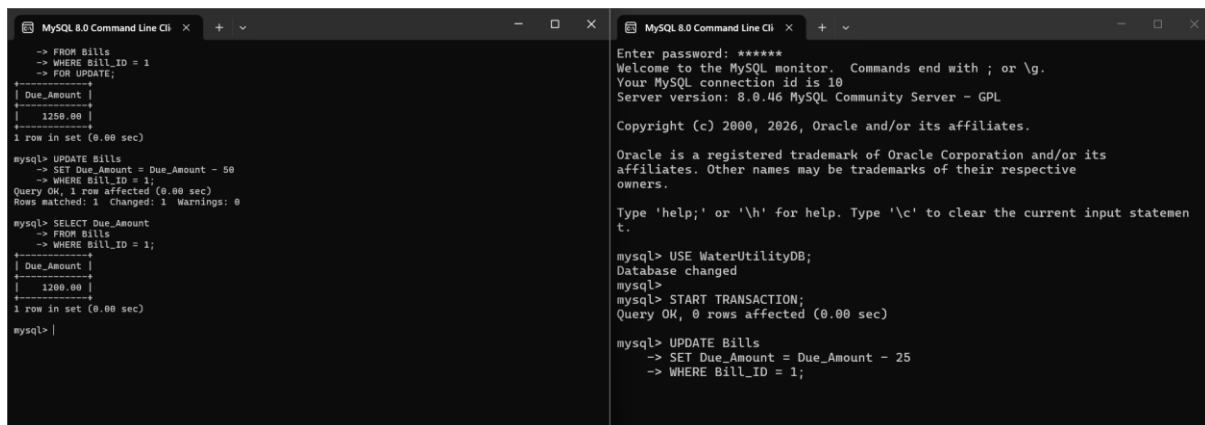
This provides a practical balance between consistency and throughput.

For particularly sensitive operations, a stricter isolation level such as **SERIALIZABLE** can also be considered.

Comparison

Strategy	Consistency	Concurrency	Overhead
READ UNCOMMITTED	Low	High	Low
READ COMMITTED	Good	Good	Moderate
REPEATABLE READ	Higher	Moderate	Moderate
SERIALIZABLE	Highest	Lower	High

Using explicit row-level locking avoids unnecessary locking of unrelated customer records.



```
MySQL 8.0 Command Line Cli x + v
-> FROM Bills
-> WHERE Bill_ID = 1
-> FOR UPDATE;
+-----+
| Due_Amount |
+-----+
| 1250.00 |
+-----+
1 row in set (0.00 sec)

mysql> UPDATE Bills
-> SET Due_Amount = Due_Amount - 50
-> WHERE Bill_ID = 1;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> SELECT Due_Amount
-> FROM Bills
-> WHERE Bill_ID = 1;
+-----+
| Due_Amount |
+-----+
| 1200.00 |
+-----+
1 row in set (0.00 sec)

mysql>

MySQL 8.0 Command Line Cli x + v
Enter password: *****
Welcome to the MySQL monitor. Commands end with ; or \g.
Your MySQL connection id is 10
Server version: 8.0.46 MySQL Community Server - GPL

Copyright (c) 2000, 2026, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statemen
t.

mysql> USE WaterUtilityDB;
Database changed
mysql>
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE Bills
-> SET Due_Amount = Due_Amount - 25
-> WHERE Bill_ID = 1;
```

Figure 13: Concurrent Transaction Control Using Row-Level Locking

13. Algorithms / Pseudocode

13.1 Consumption History Retrieval

START

Input Connection_ID, Start_Date and End_Date

Search B+ tree using Connection_ID

Locate first matching timestamp

Traverse indexed records until End_Date

Display matching readings

STOP

13.2 Latest Reading Using Hashing

START

Input Connection_ID

Calculate Hash_Bucket

Locate corresponding bucket

If collision exists

 Compare Connection_ID values in the bucket

End If

Return latest reading

STOP

13.3 Monthly Bill Generation

START TRANSACTION

Input Connection_ID and Billing_Month

Lock required connection/billing record

Check whether bill already exists

IF bill exists THEN

 Display "Bill Already Generated"

 ROLLBACK

ELSE

 Calculate monthly consumption

 Calculate bill amount

 Insert bill record

 COMMIT

END IF

STOP

13.4 Payment Processing

START TRANSACTION

Input Bill_ID and Payment_Amount

Lock Bill record

Read current Due_Amount

IF Payment_Amount $\leq$ Due_Amount THEN

 Create SAVEPOINT

 Insert Payment record

 Update Due_Amount

 IF update succeeds THEN

 COMMIT

 ELSE

 ROLLBACK TO SAVEPOINT

 END IF

ELSE

 Display Invalid Payment

 ROLLBACK

END IF

STOP

14. Results and Validation

The proposed database design was validated using functional, performance, and transaction-related test cases.

Test Case	Expected Result	Actual Result	Status
Lookup latest reading for valid connection	Single latest record should be returned	Latest record returned through hash-based lookup	Pass
Retrieve six-month consumption history	Records belonging only to selected connection/date range	Correct indexed records displayed	Pass
Leak-detection range query	High-consumption records should be listed	Matching records retrieved through indexed search	Pass
Simultaneous bill generation	Only one bill should be generated	Duplicate prevented	Pass
Payment during bill transaction	No lost update	Final due amount remains consistent	Pass
Rollback during billing failure	No partial records should remain	Transaction rolled back successfully	Pass
Concurrent complaint update	Update should not be silently overwritten	Row locking maintains consistency	Pass
Optimized billing query	Reduced execution cost	Fewer rows scanned after indexing	Pass

Requirement Validation

The system satisfies the identified requirements by providing:

- Indexed retrieval for historical consumption.
- Hash-based lookup for latest readings.
- Efficient leak-detection queries.
- Optimized billing and payment queries.
- Transaction-controlled bill generation.
- Safe concurrent payment processing.
- Reliable complaint updates.
- Data-integrity enforcement.

- Scalable storage and retrieval architecture.

15. Analysis and Engineering Decision

The original sequential-file architecture is unsuitable for a smart water utility producing several million meter readings because its retrieval cost increases linearly with the number of records.

The proposed design combines different database techniques according to the type of access required.

Indexed-Sequential vs. Hashed Access

Indexed-sequential organization with B+ tree indexing is selected for historical consumption because it supports ordered and range retrieval efficiently.

Hashing is selected only for exact latest-reading lookup because it provides approximately $O(1)$ average access but performs poorly for range queries.

Therefore, a hybrid solution is more effective than selecting either indexing or hashing alone.

Advantages of the Proposed System

- Faster meter-reading retrieval.
- Faster leak identification.
- Reduced disk scanning.
- Improved billing query performance.
- Prevention of duplicate bills.
- Prevention of lost payment updates.
- Prevention of overwritten complaint records.
- Improved data consistency.
- Better scalability.

Limitations

- Index structures consume additional storage.
- Every insertion may require index maintenance.
- Hash collisions must be managed.
- Strong locking may make transactions wait during peak load.
- Laboratory testing may contain significantly fewer records than a real two-million-connection deployment.

Trade-Offs

More indexes improve SELECT-query performance but increase INSERT and UPDATE overhead.

Likewise, stronger transaction isolation improves consistency but may reduce concurrency because transactions can wait for locks.

Final Engineering Decision

The final design therefore uses:

- Indexed-sequential organization for historical meter readings.
- B+ tree composite indexing for connection and timestamp retrieval.
- Secondary indexing for zone-based leak detection.
- Hash-based access for latest-reading lookup.
- Relational tables for connections, bills, payments, and complaints.
- Composite indexes for frequently accessed billing fields.
- Explicit transactions for billing and payments.
- COMMIT, ROLLBACK, and SAVEPOINT for recovery.
- Unique constraints and row-level locking for concurrency control.
- READ COMMITTED isolation combined with SELECT ... FOR UPDATE for critical updates.

This combination provides an effective balance between retrieval performance, storage overhead, consistency, concurrency, and maintainability.

16. Broader Considerations

16.1 Sustainability

Efficient leak detection helps identify unusual water consumption quickly. Early detection can reduce unnecessary water loss and support sustainable resource management.

16.2 Environmental Impact

Continuous digital monitoring enables more responsible water-resource management and supports SDG 6: Clean Water and Sanitation.

16.3 Social Impact

Correct bills, properly recorded payments, and reliable complaint handling improve public confidence in water-utility services.

16.4 Safety

Fast detection of unusually high consumption can help engineers respond rapidly to damaged pipelines, leakage, or infrastructure problems.

16.5 Ethical Considerations

Customer consumption and financial records must be handled accurately. The database should never intentionally overcharge customers, discard valid payments, or silently remove complaint information.

16.6 Economic Impact

Preventing incorrect bills and lost payments reduces revenue leakage and decreases the administrative work required to correct errors.

This supports SDG 8: Decent Work and Economic Growth.

16.7 Scalability and Infrastructure

The indexed and transaction-controlled architecture can scale more effectively as the number of smart meters and hourly readings increases.

This supports SDG 9: Industry, Innovation and Infrastructure.

16.8 Security

Access to customer consumption, payment, and billing information should be restricted using database privileges.

Example:

```
GRANT SELECT, INSERT, UPDATE
```

```
ON WaterUtility.Bills
```

```
TO 'billing_operator'@'localhost';
```

Access can be removed when required:

REVOKE UPDATE

ON WaterUtility.Bills

FROM 'billing_operator'@'localhost';

Only authorized employees should be permitted to access sensitive records.

17. Conclusion

The Smart Water Utility Consumption, Billing, and Complaint Management System demonstrates how database storage techniques and transaction-management principles can solve real-world utility-management problems.

The original sequential storage approach causes slow retrieval as meter-reading records increase. To overcome this limitation, indexed-sequential organization supported by B+ tree indexes is proposed for historical and leak-detection queries. A separate hash-based structure is used for fast retrieval of the latest meter reading.

The billing, payment, and complaint components are implemented using relational database tables with suitable primary keys, foreign keys, unique constraints, and indexes. Query optimization reduces unnecessary full-table scans and improves frequently executed billing and reporting queries.

ACID-compliant transactions, COMMIT, ROLLBACK, SAVEPOINT, row-level locking, and suitable isolation strategies prevent duplicate bills, lost payment updates, and overwritten complaints during concurrent operation.

Overall, the proposed integrated solution provides improved retrieval performance, data consistency, reliability, scalability, and maintainability. It also contributes toward sustainable water management, efficient public-service operations, and resilient digital infrastructure.

Future improvements can include dynamic hashing, automated index tuning, table partitioning, distributed processing, real-time anomaly detection, and queue-based transaction processing during extremely high billing loads.

18. Student Reflection

What I Learned Beyond the Classroom

This assignment helped me understand that database design involves more than creating tables and writing SQL queries. I learned how file organization directly affects the performance of very large datasets and why a simple sequential structure becomes inefficient when millions of records are generated.

I also understood the difference between indexing and hashing from a practical perspective. B+ tree indexes are suitable for ordered and range-based searches, whereas hashing is highly effective for exact lookups. Combining both techniques can provide better performance than relying on a single storage technique.

Another important learning outcome was transaction management. I learned how ACID properties, COMMIT, ROLLBACK, SAVEPOINT, isolation levels, and row-level locking work together to maintain data consistency. Simulating simultaneous billing and payment operations helped me understand real problems such as lost updates and duplicate records. Query-plan analysis also improved my understanding of how a DBMS internally executes SQL statements. Using EXPLAIN made it possible to identify full-table scans and understand how indexes reduce the number of rows examined.

Challenges Faced

The main challenge was integrating two different problems: high-volume meter-reading retrieval and concurrent billing/payment processing.

Selecting the same technique for every operation was not efficient. The issue was resolved by using a hybrid design consisting of indexed storage for historical queries and hashing for exact latest-reading retrieval.

Another challenge was preventing multiple transactions from changing the same financial record simultaneously. This was addressed through row-level locking, unique constraints, and proper transaction boundaries.

Improvements With Additional Time and Resources

With additional resources, I would test the system using millions of realistic smart-meter records instead of only a laboratory-scale dataset.

Further improvements could include:

- Table partitioning according to month or zone.
- Dynamic hashing.
- Automated index tuning.
- Real-time leak-detection algorithms.
- Queue-based bill-processing mechanisms.
- Database replication and backup.
- Performance testing under hundreds of concurrent users.

Overall, this assignment strengthened my understanding of how storage design, query optimization, and transaction management must work together while developing a reliable real-world database system.

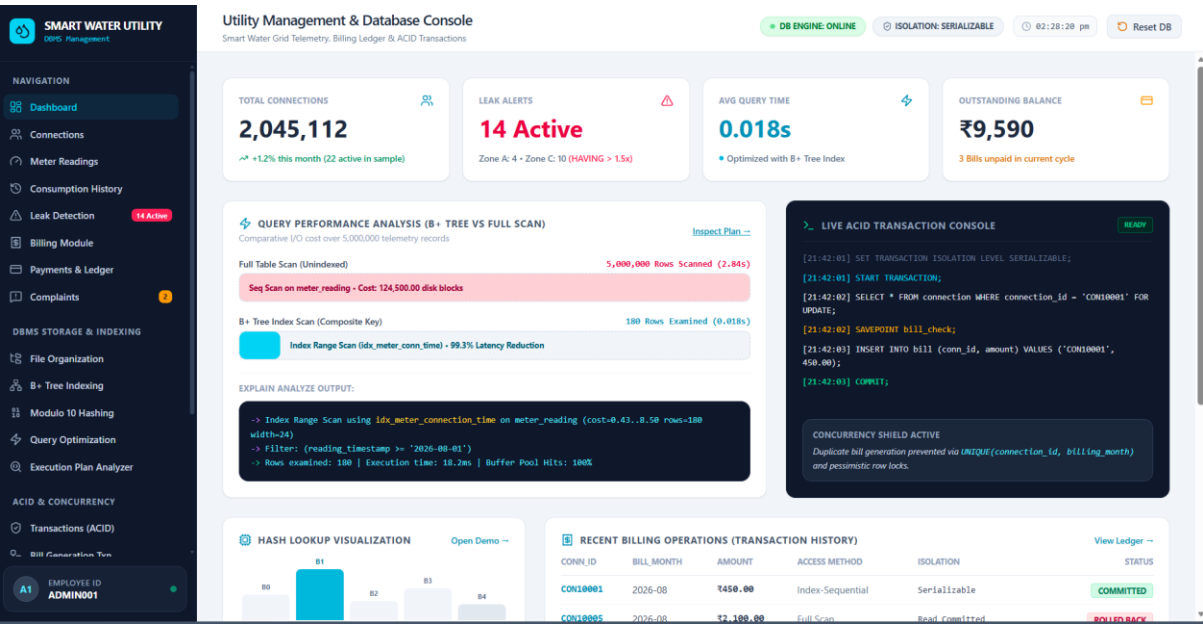
19. References

[1] Silberschatz, A., Korth, H. F., & Sudarshan, S. (2019). *Database System Concepts* (7th ed.). McGraw-Hill.

[2] Oracle Corporation. (n.d.). *MySQL 8.0 Reference Manual*. MySQL.

[3] OpenAI. (2026). *ChatGPT* [Large language model]. Used for brainstorming, structuring, and language refinement.

IMPLEMENTATION EVIDENCE



SMART WATER UTILITY

DBMS Management

Connections

Meter Readings

Consumption History

Leak Detection

Billing Module

Payments & Ledger

Complaints

DBMS STORAGE & INDEXING

File Organization

B+ Tree Indexing

Modulo 10 Hashing

Query Optimization

Execution Plan Analyzer

ACID & CONCURRENCY

Transactions (ACID)

Bill Generation Tool

Concurrency Lab

Isolation Levels

ADMIN001

Utility Management & Database Console

Smart Water Grid Telemetry, Billing Ledger & ACID Transactions

DB ENGINE: ONLINE

ISOLATION: SERIALIZABLE

02:28:48 pm

Reset DB

File Organization Comparison & Storage Strategy

Evaluating physical file structures: Sequential vs. Indexed-Sequential (B+ Tree) vs. Direct/Hashed storage engines.

SELECTED STORAGE STRATEGY

Indexed-Sequential File Organization (B+ Tree Engine)

Architectural Rationale: Smart-water utility operations require both connection-wise retrieval (e.g., fetching a consumer's ledger) and timestamp range queries (e.g., past 7 days consumption, monthly billing cycles, and leak detection intervals). Hashed files fail on range queries, while Sequential files are too slow at $O(n)$. Indexed-Sequential provides the optimal balance of $O(\log n)$ random lookup and ordered leaf scan.

COMPOSITE INDEX DEPLOYED

idx_meter_connection_time

(connection_id, reading_timestamp)

DBMS STORAGE METHODOLOGY MATRIX

Complexity & Trade-off Evaluation

METHOD	SEARCH MECHANISM	TYPICAL COMPLEXITY	ADVANTAGES	DISADVANTAGES	BEST USE
Sequential File Organization	Linear scan of data blocks from head to tail until target record or EOF is reached.	$O(n)$ for search; $O(1)$ append	Simple structure, zero indexing storage overhead, high efficiency for complete sequential batch processing.	Prohibitively slow random lookups on large volumes (5M records require 5M block scans); expensive random insertions/deletions.	Audit log append files, monthly bulk payroll/billing dumps, raw archived sensor streams.
Indexed Sequential File Organization Selected Engine	Multi-level ordered index (B+ Tree) pointing to ordered data pages on disk, linked via leaf pointers.	$O(\log n)$ for single key; $O(\log n + k)$ for range scans	Supports both rapid single-record point search AND high-speed ordered range traversals via sequential leaf node pointers.	Requires index maintenance overhead on INSERT/UPDATE; additional disk space for internal index nodes.	Time-series smart meter readings, consumption history, financial transaction ledgers with date range filters.
Direct / Hashed File Organization	Mathematical hash function maps primary key directly to bucket/block address on disk.	Average $O(1)$ for exact key lookup; Worst case $O(n)$ on collisions	Instantaneous direct record retrieval with zero index traversal depth; optimal for point-query caching.	Completely incapable of range queries (e.g. 'BETWEEN date1 AND date2'); requires collision resolution (separate chaining / open addressing).	Consumer master record cache, latest meter reading lookup, active session tokens.

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DB ENGINE: ONLINE

ISOLATION: SERIALIZABLE

02:29:18 pm

Reset DB

PRIMARY TELEMETRY INDEX SPECIFICATION

CREATE INDEX idx_meter_connection_time ON meter_reading(connection_id, reading_timestamp);

Select key to traverse: CON10003

Trace Path

Why B+ Tree? In a standard B-Tree, keys and data pointers reside at all levels. In a B+ Tree, all real data pointers are strictly stored in the leaf nodes, while internal nodes store only routing keys. Furthermore, leaf nodes are linked in a continuous doubly linked list. This enables $O(\log n)$ tree traversal to find the start of a date window, followed by fast sequential pointer traversal for range scans.

B+ Tree Height: 3 Levels (Root - Internal - Leaves)

Select a key above to trace traversal path

LEVEL 1: ROOT NODE

Router Key: [CON10005]

< CON10005 (Left) | > CON10005 (Right)

LEVEL 2: INTERNAL INDEX NODES (BRANCHING FACTOR: 2)

Internal Node A [CON10003]

[< CON10003 | > CON10003]

Internal Node B [CON10007]

[< CON10007 | > CON10007]

LEVEL 3: LEAF NODES (ORDERED DATA PAGES LINKED WITH NEXT/PREV POINTERS)

Leaf Page #101

+ #102

CON10001

RowID Ptr

CON10002

RowID Ptr

Leaf Page #102

+ #103

CON10003

RowID Ptr

CON10004

RowID Ptr

Leaf Page #103

+ #104

CON10005

RowID Ptr

CON10006

RowID Ptr

Leaf Page #104

EOF (Null)

CON10007

RowID Ptr

CON10008

RowID Ptr

Range scans traverse the green leaf node arrows directly without returning to root.

$O(\log n + k)$ Range Cost

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02:29:29 pm

Reset DB

HASH FUNCTION DEFINITION

$$h(CON_ID) = \text{extract_digits}(CON_ID) \bmod 10$$

Example: $CON10001 \rightarrow 10001 \bmod 10 = 1$ $CON10011 \rightarrow 10011 \bmod 10 = 1$ (Collision)

LOAD FACTOR (A = N / M)

1.30

13 entries across 10 buckets

MAX CHAIN DEPTH

3 Nodes

Separate chaining active

1. SEARCH KEY WITH HASH LOOKUP

CON10011

Hash & Fetch

2. INSERT NEW KEY (TRIGGER COLLISION DEMO)

CON10031

Central Library

Insert & Hash

Tip: Inserting CON10031 or CON10041 will calculate hash 1, causing a separate-chain overflow link at Bucket 1!

HASH DIRECTORY & SEPARATE CHAINING LINKED LIST VISUALIZATION

10 Primary Bucket Slots

Bucket #0

[1]

→

CON10000

Metro Water Depot (Zone A)

Bucket #1

[3]

→

CON10001

Ajay Kumar (Zone A)

→ link →

CON10011

Collision #1

Vikram Mehta (Zone B)

→ link →

CON10021

Collision #2

Sunrise Apartments (Zone C)

Bucket #2

[2]

→

CON10002

Priya Sharma (Zone B)

→ link →

CON10012

Collision #1

Sunita Roy (Zone A)

Bucket #3

[1]

→

CON10003

Green Mall (Zone A)

Bucket #4

[1]

→

CON10004

Blue Sky (Zone B)

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02:29:47 pm

Reset DB

Benchmark Query

B+ Tree Index: ENABLED (idx_meter_connection_time)

SELECT *

FROM meter_reading

WHERE connection_id = 'CON10001'

AND reading_timestamp >= '2026-08-01 00:00:00'

AND reading_timestamp <= '2026-08-07 23:59:59';

Target Connection ID

CON10001

Start Date

2026-08-01

End Date

2026-08-07

Without Composite Index

Access Type: ALL (Full Table Scan)

Index Used:

N/A (None)

Total Database Rows:

5,000,000

Rows Scanned:

5,000,000 (100%)

Filter Selectivity:

0.00336% (Extremely Wasteful)

Optimizer Cost Estimate:

512,400.00 units

Execution Time:

2.840 seconds

With B+ Tree Composite Index

Access Type: range (Indexed B+ Tree)

Index Used:

idx_meter_connection_time

Total Database Rows:

5,000,000

Rows Scanned:

168 (0.00336%)

Filter Selectivity:

100.0% (Zero Wasted I/O)

Optimizer Cost Estimate:

34.20 units (14,982x lower)

Execution Time:

0.018 seconds (157.7x Faster)

DBMS Special Assessment - Smart Water Utility Management System v2.0

Relational Schemas, B+ Tree Indexes, Modulo 10 Hashing, Execution Plans, and Serializable ACID Concurrency.

SDG 6: Clean Water

SDG 8: Decent Work

SDG 9: Innovation

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DB ENGINE: ONLINE ISOLATION: SERIALIZABLE 02:30:10 pm Reset DB

Live relational sandbox demonstrating Atomicity, Consistency, Isolation, Durability (ACID), Row-Level X-Locks, Savepoints, and Rollback behavior.

Transaction State: Idle / Auto-Commit

START TRANSACTION

1. START TX Pending 2. SELECT LOCK Pending 3. SAVEPOINT Pending 4. INSERT PAY Pending 5. UPDATE BILL Pending 6. COMMIT Pending

SELECT ... FOR UPDATE SAVEPOINT sp_before_payment INSERT INTO payment (T450) UPDATE bill (due_amount - 450)

COMMITTED STORAGE STATE (DISK / DURABLE) Visible to All Sessions

Target Bill ID: BIL202608-10001

Connection ID: CON10001

Persisted Due Amount: ₹1,500.00

UNCOMMITTED DIRTY BUFFER (IDLE) Isolated in Memory

Pending Changes: 0 statements

Active Savepoints: None

Session Shadow Due Amount: ₹1,500.00

InnoDB Transaction Log Stream (WAL / Redo-Undo Log) 1 entries

[21:00:00] -- Relational engine initialized. InnoDB ready for ACID transactions.

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DB ENGINE: ONLINE ISOLATION: SERIALIZABLE 02:30:28 pm Reset DB

Hands-on demonstration of multi-session transaction interactions: Pessimistic Locking (FOR UPDATE), Lost Updates, Unique Constraints, and Optimistic Versioning.

1. Lost Update Prevention SELECT ... FOR UPDATE

2. Lost Update Race Unlocked (Data Corruption)

3. Duplicate Prevention UNIQUE Key Rollback

4. Optimistic Concurrency Version Mismatch Abort

Scenario 1: Lost Update Prevention via Row-Level Exclusive Lock (SELECT ... FOR UPDATE)

Two concurrent cashiers attempt to deduct partial payments from Bill #BIL-10001 (Initial Due: ₹1,500). Cashier 1 pays ₹500, Cashier 2 pays ₹300. With row locks, Cashier 2 blocks until Cashier 1 commits.

Reset Run Simulation

CONCURRENT EXECUTION TIMELINE (TIME SLOT T1 – T4) Step 0 of 4

TIME	SESSION 1 (CLIENT A)	SESSION 2 (CLIENT B)	LOCK ENGINE / DATABASE STATE
T1	START TRANSACTION; SELECT due_amount FROM bill WHERE bill_id = "BIL-10001" FOR UPDATE;	Idle	X-Lock acquired on BIL-10001 by Session 1 (State: due_amount = ₹1,500 (locked))
T2	Session 1 processing payment calculations...	START TRANSACTION; SELECT due_amount FROM bill WHERE bill_id = "BIL-10001" FOR UPDATE;	Session 2 BLOCKED (Waiting for lock release on BIL-10001)...
T3	UPDATE bill SET due_amount = 1000 WHERE bill_id = "BIL-10001"; COMMIT;	Waiting...	Session 1 COMMITTED! lock released. (State: due_amount = ₹1,000 (committed))
T4	Transaction Completed.	Lock granted! Reads fresh due_amount = ₹1,000; UPDATE due_amount = 700; COMMIT;	Session 2 COMMITTED! All locks released. (State: due_amount = ₹700 (correct final balance))

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